

Stelios Zacharioudakis

MACHINE LEARNING RESEARCH & ENGINEERING

stelios@stelioszach.com | stelioszach.com | github.com/stelioszach03
linkedin.com/in/stelios-zach

Machine learning research and software engineering, focused on generative models, model-based reinforcement learning, reliable evaluation and deployed systems.

ENGINEERING EXPERIENCE

Former Head Engineer - Paphos Medical Association | Pro bono

Jun 2022 - Jul 2026

- Led pro bono engineering for AsklepiosMed, a donated member-services platform for the association.
- Developed full-stack web applications and administrative workflows using React, Node.js/Express and PostgreSQL.
- Maintained the platform and supported its deployment and operation on a Linux VPS.

RESEARCH

World-model learning and evaluation | Independent research, ongoing and unpublished

- Investigate imagination horizons, critic stability and policy selection with PyTorch latent world models and imagined actor-critic learning.
- Design controlled ablations, paired simulator evaluations and traceable analyses to study task-dependent outcomes and failed hypotheses.

Score-based generative models for undersampled MRI reconstruction | BSc thesis

- Reimplemented a published reconstruction method in PyTorch using the authors' pretrained prior; developed MRI measurement operators and a Flax-to-PyTorch checkpoint conversion.
- The thesis reports reproduction within 0.15 dB PSNR on 256 BraTS slices at 4x, 8x and 24x sampling factors; evaluates Monte Carlo uncertainty and retrospective domain shift.

Thesis document: September 2026. Reproduction and experimental analysis; no claim of clinical validation.

SELECTED SOFTWARE PROJECTS

Constraint verification with Z3 - Structured candidate checking, explicit solver outcomes and a live verification demo.

Speculative decoding lab - Recorded inference experiments, baseline comparisons and reproducibility notes, including slower configurations.

Applied ML demos - Text de-identification, synthetic graph-based transaction scoring and public transit anomaly exploration; prototype scope and limitations documented.

EDUCATION

BSc in Computer Science - National and Kapodistrian University of Athens

Department of Informatics and Telecommunications | June 2026

TECHNICAL FOCUS

ML: Python, PyTorch, NumPy, SciPy, scikit-learn; generative models, model-based RL, uncertainty analysis and experimental evaluation.

Systems: FastAPI, Node.js/Express, React, JavaScript, SQL/PostgreSQL, Redis, Linux, Nginx, Docker, Git and automated testing.